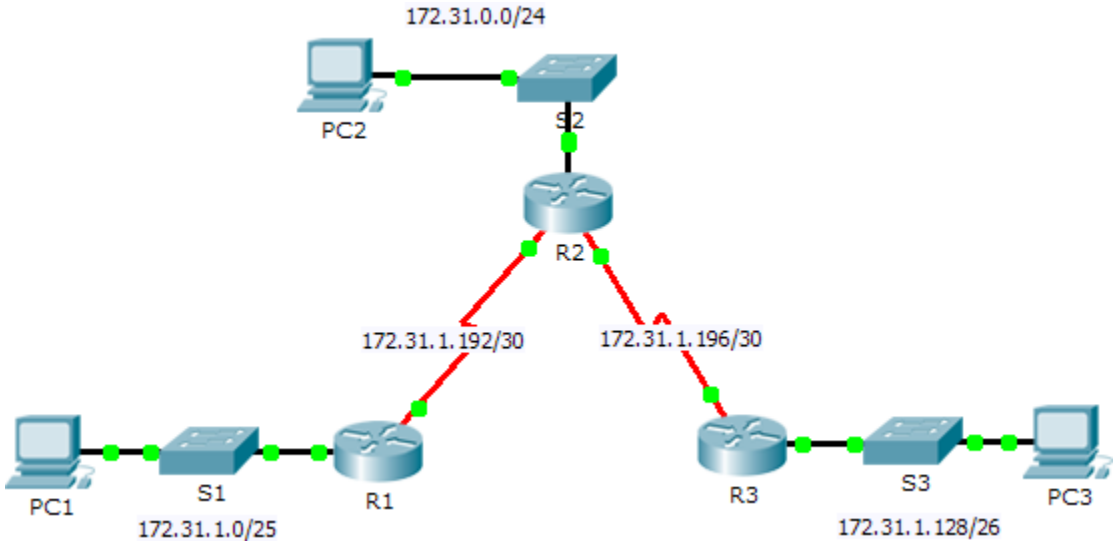
Packet Tracer - Configuring IPv4 Static and Default Routes

Topology



Addressing Table

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
R1	G0/0	172.31.1.1	255.255.255.128	N/A
	S0/0/0	172.31.1.194	255.255.255.252	N/A
R2	G0/0	172.31.0.1	255.255.255.0	N/A
	S0/0/0	172.31.1.193	255.255.255.252	N/A
	S0/0/1	172.31.1.197	255.255.255.252	N/A
R3	G0/0	172.31.1.129	255.255.255.192	N/A
	S0/0/1	172.31.1.198	255.255.255.252	N/A
PC1	NIC	172.31.1.126	255.255.255.128	172.31.1.1
PC2	NIC	172.31.0.254	255.255.255.0	172.31.0.1
PC3	NIC	172.31.1.190	255.255.255.192	172.31.1.129

Objectives

Part 1: Examine the Network and Evaluate the Need for Static Routing

Part 2: Configure Static and Default Routes

Part 3: Verify Connectivity

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Background

In this activity, you will configure static and default routes. A static route is a route that is entered manually by the network administrator to create a reliable and safe route. There are four different static routes that are used in this activity: a recursive static route, a directly attached static route, a fully specified static route, and a default route.

Part 1: Examine the Network and Evaluate the Need for Static Routing

a. Looking at the topology diagram, how many networks are there in total? 5

b. How many networks are directly connected to R1, R2, and R3? 2 + 3 + 2 = 7

c. How many static routes are required by each router to reach networks that are not directly connected?

R1: 3 static routes, R2: 2 static routes, R3: 3 static routes (or 1 default route)

d. Test connectivity to the R2 and R3 LANs by pinging PC2 and PC3 from PC1.

Why were you unsuccessful? The routers have no routing table entries for remote destination networks, so packets are dropped at the first gateway.

Part 2: Configure Static and Default Routes

Step 1: Configure recursive static routes on R1.

a. What is recursive static route?
A recursive static route is a static route that specifies only the next-hop IPv4 address (not the exit interface).

b. Why does a recursive static route require two routing table lookups?
The router performs the first lookup to match the destination network and find the next-hop IP address. It then performs a second lookup to resolve that next-hop IP address to an outbound exit interface.

c. Configure a recursive static route to every network not directly connected to R1, including the WAN link between R2 and R3.

```
R1(config)# ip route 172.31.0.0 255.255.255.0 172.31.1.193
R1(config)# ip route 172.31.1.196 255.255.255.252 172.31.1.193
R1(config)# ip route 172.31.1.128 255.255.255.192 172.31.1.193
```

d. Test connectivity to the R2 LAN and ping the IP addresses of PC2 and PC3.

Why were you unsuccessful?

R2 and R3 do not yet have routing table entries (return routes) for R1's LAN (`172.31.1.0/25`). Although the echo request packets reach the destination devices, the reply packets are dropped by R2 or R3 because they have no route back to PC1.

Step 2: Configure directly attached static routes on R2.

a. How does a directly attached static route differ from a recursive static route? A directly attached static route specifies the local exit interface instead of the next-hop IP address, requiring only a single routing table lookup.

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b. Configure a directly attached static route from R2 to every network not directly connected.
```
R2(config)# ip route 172.31.1.0 255.255.255.128 Serial0/0/0
R2(config)# ip route 172.31.1.128 255.255.255.192 Serial0/0/1
```

c. Which command only displays directly connected networks? `show ip route connected`

d. Which command only displays the static routes listed in the routing table? `show ip route static`

e. When viewing the entire routing table, how can you distinguish between a directly attached static route and a directly connected network?
Directly connected networks are prefixed with the **C** code, whereas directly attached static routes are prefixed with the **S** code and display the text *"is directly connected, [Interface]"*.

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Step 3: Configure a default route on R3.

a. How does a default route differ from a regular static route?
A default route uses the quad-zero mask `0.0.0.0 0.0.0.0` to match all packets that do not match any other specific route in the routing table (acting as the Gateway of Last Resort), whereas a regular static route matches only a specific network destination and subnet mask.

b. Configure a default route on R3 so that every network not directly connected is reachable.
```
R3(config)# ip route 0.0.0.0 0.0.0.0 Serial0/0/1
```

c. How is a static route displayed in the routing table? A standard static route is displayed with the **S** code, and a default static route is displayed with the **S*** code (*indicates a candidate default route).

Step 4: Document the commands for fully specified routes.

Note: Packet Tracer does not currently support configuring fully specified static routes. Therefore, in this step, document the configuration for fully specified routes.

a. Explain a fully specified route.
A fully specified static route specifies both the outbound exit interface and the next-hop IPv4 address within the same command.

b. Which command provides a fully specified static route from R3 to the R2 LAN?
```
ip route 172.31.0.0 255.255.255.0 Serial0/0/1 172.31.1.197
```

c. Write a fully specified route from R3 to the network between R2 and R1. Do not configure the route; just calculate it.
```
ip route 172.31.1.192 255.255.255.252 Serial0/0/1 172.31.1.197
```

d. Write a fully specified static route from R3 to the R1 LAN. Do not configure the route; just calculate it.
```
ip route 172.31.1.0 255.255.255.128 Serial0/0/1 172.31.1.197
```

Step 5: Verify static route configurations.

Use the appropriate **show** commands to verify correct configurations.

Which **show** commands can you use to verify that the static routes are configured correctly?

`show ip route` or `show ip route static`
`show running-config` (or `show run | section ip route`)

Part 3: Verify Connectivity

Every device should now be able to ping every other device. If not, review your static and default route configurations.

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Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 1: Examine the Network and Evaluate the Need for Static Routing	a - d	10	
	Part 1 Total	**10**	
Part 2: Configure Static and Default Routes	Step 1	7	
	Step 2	7	
	Step 3	3	
	Step 4	10	
	Step 5	3	
	Part 2 Total	**30**	
	Packet Tracer Score	**60**	
	Total Score	**100**	

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